Niranjana Kulkarni

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Education

Texas A&M University — College Station, TX
Ph.D. in Mathematics — *Aug 2025 – Present*

Texas A&M University — College Station, TX
M.S. in Applied Mathematics (Computational Mathematics) — *Jan 2023 – Dec 2024*

- Thesis: "On the Radial Reduction of the Linearization at a Solitary Wave in the Nonlinear Dirac Equation in Three Spatial Dimensions."

University of California, Davis — Davis, CA
B.S. in Applied Mathematics — *Jun 2020 – Jun 2022*

Technical Skills

- **Programming:** Python, R, MATLAB, AMPL, JavaScript; **Libraries:** NumPy, SciPy, pandas, Matplotlib
- **Tools & Computing:** LaTeX, Jupyter Notebook, RStudio, Unix, SLURM (Texas A&M Grace HPC cluster), Excel

Research Experience

Partial Differential Equations | *with Dr. Andrew Comech, Texas A&M University* — Jan 2024 – Present

- Investigating the spectral stability of one-frequency and bi-frequency solitary waves in the nonlinear Dirac equation.
- Automating large-scale computations and data visualization on a high-performance computing cluster.
- Analyzing radial reductions and spectral properties of linearized differential operators.

Directed Studies Course in Knot Theory | *Dr. Jennifer Schultens, UC Davis* — Jun 2022 – Sep 2022

- Completed faculty-guided independent reading in knot theory and its mathematical applications.

Publications & Presentations

- **Journal article (2025):** A. Comech, **N. Kulkarni**, N. Boussaïd, and J. Cuevas-Maraver. "Stable bi-frequency spinor modes as Dark Matter candidates." *Journal of High Energy Physics*, article 82.
- **Preprint (2024):** N. Boussaïd, A. Comech, and **N. Kulkarni**. "On spectral stability of one- and bi-frequency solitary waves in Soler model in (3+1)D." arXiv:2412.21170.
- **Presentation:** Invited research talk on the nonlinear Dirac equation at Texas A&M PDEs Working Seminar.

Selected Honors & Awards

- **Dean's Honors List** (Dec 2020, Dec 2021, Mar 2022, Jun 2022), College of Letters & Science, UC Davis.

Teaching Experience

Graduate Teaching Assistant | *Texas A&M University* — Aug 2025 – Present

- Led recitation sessions, office hours; provided technical instruction in calculus, numerical analysis, linear algebra, and computational methods, including Python; graded homework and assessments with structured rubrics.

Graduate Student Assistant Grader | *Texas A&M University* — Aug 2023 – Dec 2023

- Graded homework in Complex Analysis as well as Combinatorics, providing feedback on mathematical reasoning.

Additional Experience

Mathematics and Statistics Fair Volunteer | *Texas A&M University* — Apr 2023

- Designed lesson materials and example problems, and provided one-on-one mathematics instruction to visiting high-school students.

Application of Computers in Technology Intern | *UC Davis* — Jan 2022 – Jun 2022

- Taught R, statistics, and Excel to **100+ students**; developed curriculum for **300+ students** and graded work for 40+ students weekly.